

LOCALIZING MOTIVES OVER NONZERO CONNECTIVE $\mathbb{E}_2$ -RINGS ARE NOT COMPACTLY GENERATED

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ABSTRACT. We prove that $\mathrm{Mot}_R^{\mathrm{loc}}$ is not compactly generated for every nonzero connective $\mathbb{E}_2$ -ring spectrum R , removing the nonzero-rationalization hypothesis from Ramzi’s theorem. The new input is the positive-characteristic case. For a perfect field k of characteristic p , we construct a k -linear motive whose cyclotomic topological Hochschild homology is compact but not dualizable over $\mathrm{THH}(k)$. The compact object comes from the one-weight polygonic cell, and its nondualizability is detected by relative Tate assembly. Ramzi’s universal THH -motive realizes the cell, while Efimov’s rigidity theorem converts the discrepancy into failure of compact generation. The general case follows by passage to a perfect residue field, together with Ramzi’s rational result.

1. INTRODUCTION

For an $\mathbb{E}_2$ -ring spectrum R , write $\mathrm{Mot}_R^{\mathrm{loc}}$ for the presentable stable ∞ -category of R -linear localizing motives. When R is commutative, this is the category of localizing motives relative to the rigid small base $\mathrm{Perf}(R)$ in the sense of Hoyo–Scherotzke–Sibilla [HSS17, Section 5.3]; in Efimov’s presentable-base formalism the corresponding base is Mod_R .

Ramzi proved that $\mathrm{Mot}_R^{\mathrm{loc}}$ is not compactly generated when R is a connective $\mathbb{E}_2$ -ring spectrum with nonzero rationalization. He observed that his method does not cover ring spectra supported at a prime, such as $\mathbb{F}_p$ -algebras [Ram26, Introduction]. The purpose of this paper is to supply the missing positive-characteristic argument and then remove the rationalization hypothesis.

Theorem 1.1 (Main theorem). *Let R be a nonzero connective $\mathbb{E}_2$ -ring spectrum. Then $\mathrm{Mot}_R^{\mathrm{loc}}$ is not compactly generated.*

The new part of the theorem is the following more precise statement. We regard an ordinary field as its Eilenberg–MacLane $\mathbb{E}_\infty$ -ring spectrum.

Theorem 1.2 (Perfect fields of positive characteristic). *Let k be a perfect field of characteristic $p > 0$. There is an object $M_k \in \mathrm{Mot}_k^{\mathrm{loc}}$ such that*

$$M_k \notin \mathrm{Loc}_{\mathrm{Mot}_k^{\mathrm{loc}}}((\mathrm{Mot}_k^{\mathrm{loc}})^\omega).$$

In particular, $\mathrm{Mot}_k^{\mathrm{loc}}$ is not compactly generated.

The proof has four inputs.

- (i) Efimov’s rigidity theorem identifies compact objects of $\mathrm{Mot}_k^{\mathrm{loc}}$ with dualizable objects [Ef25, Theorem 0.3]; in the notation used here, this consequence is stated explicitly in [Ram26, Theorem 1.1].
- (ii) The one-weight induction functor of Krause–McCandless–Nikolaus produces a compact cyclotomic spectrum C whose underlying $\mathbb{T}$ -spectrum is $\Omega\Sigma_+^\infty\mathbb{T}$ and whose cyclotomic Frobenius maps are zero [KMN23, Example 2.7, Construction 2.20, Proposition 2.21].

- (iii) If $A = \mathrm{THH}(k)$, then the free cyclotomic A -module $A \otimes C$ is not dualizable. Its relative Tate assembly map has nonzero source and zero target, contradicting the dualizability criterion of Kondyrev–Mazel–Gee–Shah [KMGS21, Example 4.11].
- (iv) Ramzi’s universal THH -motive produces an absolute localizing motive M_0 with $\mathrm{THH}(M_0) \simeq \Omega\Sigma_+^\infty \mathbb{T}$ equivariantly for the intrinsic circle action [Ram25, Corollary 2.2] and [Ram26, Remark 2.4]. Since this circle spectrum is Tate-acyclic for every finite cyclic subgroup, the equivalence has a unique promotion to cyclotomic spectra.

The remaining argument for [Theorem 1.2](#) is formal: base-change M_0 to k , apply relative cyclotomic THH , and use compactness to split the resulting object off the THH of one compact motive. Symmetric monoidality would then force that object to be dualizable, giving a contradiction.

To deduce [Theorem 1.1](#), let R be nonzero and connective. The case $R_{\mathbb{Q}} \neq 0$ is Ramzi’s theorem. Otherwise a perfect closure of a residue field of $\pi_0 R$ has positive characteristic. Base-change takes the distinguished witness over R to the witness of [Theorem 1.2](#); because this base-change is an internal left adjoint, it preserves compact objects and their localizing envelope. This residue-field step is carried out in [section 6](#).

2. CONVENTIONS AND CATEGORICAL PRELIMINARIES

Until [section 6](#), fix a prime p and a perfect field k of characteristic p , regarded as the ring spectrum Hk .

2.1. Integral and one-prime cyclotomic spectra. Write $\mathbb{T} = S^1$. We use the Nikolaus–Scholze category CycSp of integral cyclotomic spectra. Thus an object has an underlying object of $\mathrm{Sp}^{B\mathbb{T}}$ and, for every prime ℓ , a $\mathbb{T}$ -equivariant cyclotomic Frobenius

$$\varphi_\ell: X \longrightarrow X^{tC_\ell},$$

where the residual $\mathbb{T}/C_\ell$ -action is identified with a $\mathbb{T}$ -action by the ℓ th-root isomorphism; see [NS18, Definition II.1.1].

We shall use a one-prime category that retains the *full* circle action. Define

$$\mathrm{CycSp}_{\mathbb{T}}^{(p)} := \mathrm{LEq}_{\mathrm{Sp}^{B\mathbb{T}}}(\mathrm{id}, (-)^{tC_p}). \quad (2.1)$$

An object is a circle spectrum X together with one map $X \rightarrow X^{tC_p}$. The lax symmetric monoidality of the Tate construction equips (2.1) with a symmetric monoidal structure [KMGS21, Example 4.11]. Forgetting the Frobenius maps at primes $\ell \neq p$ defines a symmetric monoidal projection

$$\rho_p: \mathrm{CycSp} \longrightarrow \mathrm{CycSp}_{\mathbb{T}}^{(p)}. \quad (2.2)$$

This is not the Nikolaus–Scholze p -typical category whose underlying action is only by C_{p^∞} ; the distinction is important because the criterion used below is formulated in the full-circle lax equalizer (2.1).

For a presentably symmetric monoidal stable ∞ -category $\mathcal{C}$ and a commutative algebra $B \in \mathrm{CAlg}(\mathcal{C})$, the category $\mathrm{Mod}_B(\mathcal{C})$ is equipped with its relative tensor product. The free-module functor

$$B \otimes -: \mathcal{C} \longrightarrow \mathrm{Mod}_B(\mathcal{C})$$

is colimit-preserving and symmetric monoidal, and its right adjoint is the forgetful functor, which creates colimits.

An object is called *compact* if it is ω -compact. For a collection of objects S in a stable presentable ∞ -category $\mathcal{C}$, we write $\mathrm{Loc}_{\mathcal{C}}(S)$ for the smallest full stable subcategory of $\mathcal{C}$ that contains S and is closed under all small colimits.

2.2. A formal obstruction. The following fact makes the final retract argument explicit.

Lemma 2.1 (Compact envelope). *Let $\mathcal{C}$ be a presentable stable ∞ -category and let $\mathcal{C}^\omega$ be its essentially small full subcategory of compact objects. The canonical functor*

$$j: \operatorname{Ind}(\mathcal{C}^\omega) \longrightarrow \mathcal{C}$$

extending the inclusion of $\mathcal{C}^\omega$ is fully faithful, and its essential image is $\operatorname{Loc}_{\mathcal{C}}(\mathcal{C}^\omega)$. Consequently, every object of $\operatorname{Loc}_{\mathcal{C}}(\mathcal{C}^\omega)$ is a filtered colimit, computed in $\mathcal{C}$, of objects of $\mathcal{C}^\omega$.

Proof. The inclusion $\mathcal{C}^\omega \hookrightarrow \mathcal{C}$ is exact. Its Ind-extension j preserves filtered colimits and finite colimits. It also preserves arbitrary coproducts: a coproduct is the filtered colimit, over the finite subsets of its indexing set, of the corresponding finite coproducts. An exact functor between stable ∞ -categories that preserves arbitrary coproducts preserves all small colimits. Thus j is colimit-preserving.

If $X \simeq \operatorname{colim}_i X_i$ and $Y \simeq \operatorname{colim}_a Y_a$ are filtered colimits of objects of $\mathcal{C}^\omega$, then compactness of every X_i gives

$$\begin{aligned} \operatorname{Map}_{\mathcal{C}}(jX, jY) &\simeq \lim_i \operatorname{Map}_{\mathcal{C}}(X_i, \operatorname{colim}_a Y_a) \\ &\simeq \lim_i \operatorname{colim}_a \operatorname{Map}_{\mathcal{C}}(X_i, Y_a), \end{aligned}$$

which is the mapping-space formula in $\operatorname{Ind}(\mathcal{C}^\omega)$. Thus j is fully faithful.

Its essential image is closed under all colimits and contains $\mathcal{C}^\omega$. Hence it contains $\operatorname{Loc}_{\mathcal{C}}(\mathcal{C}^\omega)$. Conversely, every object of an Ind-completion is a filtered colimit of representables; therefore the essential image of j is contained in the localizing subcategory generated by $\mathcal{C}^\omega$. This proves both assertions. $\square$

Lemma 2.2 (Retracts of dualizable objects). *Let $\mathcal{D}$ be an idempotent-complete symmetric monoidal ∞ -category. The full subcategory $\mathcal{D}^{\operatorname{dual}}$ of dualizable objects is closed under retracts.*

Proof. Let X be a retract of a dualizable object Y , represented by an idempotent $e: Y \rightarrow Y$. The contravariant duality functor on $\mathcal{D}^{\operatorname{dual}}$ sends e to an idempotent $e^\vee: Y^\vee \rightarrow Y^\vee$. Since $\mathcal{D}$ is idempotent complete, split $e^\vee$ and denote its image by $X^\vee$. Composing the evaluation and coevaluation of Y with the two splittings gives maps

$$X^\vee \otimes X \longrightarrow \mathbf{1}, \quad \mathbf{1} \longrightarrow X \otimes X^\vee.$$

The triangle identities follow by applying the splittings to the triangle identities for Y . Hence X is dualizable. $\square$

Proposition 2.3 (Compact nondualizable image obstruction). *Let $\mathcal{C}$ and $\mathcal{D}$ be presentable stable symmetric monoidal ∞ -categories, and let*

$$F: \mathcal{C} \longrightarrow \mathcal{D}$$

be a colimit-preserving symmetric monoidal functor. Assume

$$\mathcal{C}^\omega = \mathcal{C}^{\operatorname{dual}}.$$

If there is an object $M \in \mathcal{C}$ such that $F(M)$ is compact but not dualizable in $\mathcal{D}$, then

$$M \notin \operatorname{Loc}_{\mathcal{C}}(\mathcal{C}^\omega).$$

In particular, $\mathcal{C}$ is not compactly generated.

Proof. Suppose that $M \in \operatorname{Loc}_{\mathcal{C}}(\mathcal{C}^\omega)$. By Lemma 2.1, there is a filtered diagram (M_i) in $\mathcal{C}^\omega$ and an equivalence

$$M \simeq \operatorname{colim}_i M_i.$$

Since F preserves colimits,

$$F(M) \simeq \operatorname{colim}_i F(M_i).$$

Compactness of $F(M)$ implies that the identity map of $F(M)$ factors through one stage: there are maps

$$F(M) \xrightarrow{u} F(M_i) \xrightarrow{v} F(M) \quad \text{with} \quad v \circ u \simeq \operatorname{id}_{F(M)}.$$

Thus $F(M)$ is a retract of $F(M_i)$. The object M_i is dualizable by the hypothesis $\mathcal{C}^\omega = \mathcal{C}^{\text{dual}}$, so $F(M_i)$ is dualizable because F is symmetric monoidal. By [Lemma 2.2](#), $F(M)$ is dualizable, a contradiction. $\square$

3. A COMPACT NONDUALIZABLE CYCLOTOMIC $\operatorname{THH}(k)$ -MODULE

Set

$$A = \operatorname{THH}(k) \in \operatorname{CAlg}(\operatorname{CycSp}).$$

We construct an object

$$Q \in \operatorname{Mod}_A(\operatorname{CycSp})$$

that is compact but not dualizable.

3.1. The one-weight cyclotomic cell. A truncation set is a subset $T \subseteq \mathbb{N}_{>0}$ closed under divisors. Let $\langle 1 \rangle = \{1\}$. Krause–McCandless–Nikolaus define the category PgcSp_T of T -typical polygonic spectra and a functor

$$i_T : \operatorname{CycSp} \longrightarrow \operatorname{PgcSp}_T.$$

For $T = \langle 1 \rangle$, evaluation at the unique weight is an equivalence

$$\operatorname{PgcSp}_{\langle 1 \rangle} \simeq \operatorname{Sp}$$

by [\[KM23, Example 2.7\]](#). Moreover, i_T admits a left adjoint L_T by [\[KM23, Proposition 2.21 and Theorem 2.24\]](#). Under the displayed equivalence, Construction 2.20 of [\[KM23\]](#) gives

$$U(L_{\langle 1 \rangle}(X)) \simeq \operatorname{ind}_1(X) \simeq \Sigma_+^\infty \mathbb{T} \otimes X \quad \text{in } \operatorname{Sp}^{B\mathbb{T}},$$

where $\mathbb{T}$ acts by translation. Since $\langle 1 \rangle / \ell = \emptyset$ for every prime ℓ , all cyclotomic Frobenius maps on $L_{\langle 1 \rangle}(X)$ are zero.

Define

$$E = L_{\langle 1 \rangle}(\mathbb{S}), \quad C = \Omega E. \tag{3.1}$$

Lemma 3.1 (The one-weight cell). *The cyclotomic spectra E and C are compact. Their underlying $\mathbb{T}$ -spectra are*

$$U(E) \simeq \operatorname{ind}_1(\mathbb{S}) \simeq \Sigma_+^\infty \mathbb{T}, \quad U(C) \simeq \Omega \Sigma_+^\infty \mathbb{T},$$

and every cyclotomic Frobenius on either object is zero. Equivariant Atiyah duality also gives

$$U(C) \simeq \operatorname{coind}_1(\mathbb{S}) = F(\mathbb{T}_+, \mathbb{S}).$$

Proof. The functor i_T preserves all small colimits; this is stated in the proof of [\[KM23, Theorem 2.24\]](#), where the colimits are computed on the underlying objects. Hence, for every filtered diagram (Y_a) in CycSp ,

$$\begin{aligned} \operatorname{Map}_{\operatorname{CycSp}}(E, \operatorname{colim}_a Y_a) &\simeq \operatorname{Map}_{\operatorname{Sp}}(\mathbb{S}, i_{\langle 1 \rangle} \operatorname{colim}_a Y_a) \\ &\simeq \operatorname{Map}_{\operatorname{Sp}}(\mathbb{S}, \operatorname{colim}_a i_{\langle 1 \rangle} Y_a) \\ &\simeq \operatorname{colim}_a \operatorname{Map}_{\operatorname{CycSp}}(E, Y_a). \end{aligned}$$

Thus E is compact. Since CycSp is stable, its desuspension $C = \Omega E$ is compact as well. The formulas for the underlying objects and the vanishing of the Frobenius maps follow directly from [KMN23, Construction 2.20].

Finally, equivariant Atiyah duality for the compact Lie group $\mathbb{T}$ identifies coinduction from the trivial subgroup with the one-fold desuspension of induction:

$$\text{coind}_1(\mathbb{S}) \simeq \Omega \text{ind}_1(\mathbb{S}).$$

This is also the identification used in the proof of [KMN23, Theorem 2.24]. $\square$

Lemma 3.2 (Finite-subgroup Tate acyclicity). *For every prime ℓ ,*

$$U(C)^{tC_\ell} \simeq 0.$$

More generally, for every $Y \in \text{Sp}^{BC_\ell}$,

$$(Y \otimes U(C)|_{C_\ell})^{tC_\ell} \simeq 0.$$

Proof. Under translation, $\mathbb{T}$ is a finite free C_ℓ -CW complex: a CW structure on the quotient $\mathbb{T}/C_\ell \simeq \mathbb{T}$ lifts to one free orbit of 0-cells and one free orbit of 1-cells. Consequently, $\Sigma_+^\infty \mathbb{T}|_{C_\ell}$ belongs to the stable subcategory $\text{Sp}_{\text{ind}}^{BC_\ell}$ generated by spectra induced from the trivial subgroup. The same is true after desuspension. Nikolaus–Scholze prove that the Tate construction vanishes on $\text{Sp}_{\text{ind}}^{BC_\ell}$ and that this subcategory is a tensor ideal [NS18, Lemma I.3.8(i)–(ii)]. Applying these two assertions gives both claims. $\square$

3.2. The free A -module. Define

$$Q = A \otimes C \in \text{Mod}_A(\text{CycSp}). \quad (3.2)$$

Lemma 3.3 (Compactness of the free module). *The object Q is compact in $\text{Mod}_A(\text{CycSp})$.*

Proof. For a filtered diagram (N_i) of cyclotomic A -modules, the free-forgetful adjunction and Lemma 3.1 give

$$\begin{aligned} \text{Map}_{\text{Mod}_A(\text{CycSp})} \left(A \otimes C, \text{colim}_i N_i \right) &\simeq \text{Map}_{\text{CycSp}} \left(C, U \text{colim}_i N_i \right) \\ &\simeq \text{Map}_{\text{CycSp}} \left(C, \text{colim}_i U N_i \right) \\ &\simeq \text{colim}_i \text{Map}_{\text{Mod}_A(\text{CycSp})} (A \otimes C, N_i). \end{aligned}$$

Here the forgetful functor creates colimits. Thus Q is compact. $\square$

Proposition 3.4 (Failure of dualizability). *The compact integral cyclotomic A -module Q is not dualizable.*

Proof. The projection ρ_p in (2.2) is symmetric monoidal. At the level of modules, the symmetric monoidal lax-limit description of [KMGS21, Remark 1.6] gives a strong symmetric monoidal functor

$$\rho_p^A: \text{Mod}_A(\text{CycSp}) \longrightarrow \text{Mod}_{\rho_p A}(\text{CycSp}_{\mathbb{T}}^{(p)}).$$

We abbreviate $\rho_p^A(Q)$ to $\rho_p Q$. Thus, if Q were dualizable, then its image

$$\rho_p Q \in \text{Mod}_{\rho_p A}(\text{CycSp}_{\mathbb{T}}^{(p)})$$

would be dualizable as well. We show that it is not.

For readability, continue to write A for the underlying circle-equivariant algebra $\rho_p A$, and let $X = U(\rho_p Q)$ be the underlying circle-equivariant A -module. The Frobenius $\varphi_A: A \rightarrow A^{tC_p}$ makes A^{tC_p} an A -algebra. The Frobenius of X is φ_A -semilinear and hence has an adjoint

$$\tilde{\varphi}_X: X \otimes_{A, \varphi_A} A^{tC_p} \longrightarrow X^{tC_p}. \quad (3.3)$$

For a perfect field of characteristic p , Kondyrev–Mazel–Gee–Shah prove that a $\mathrm{THH}(k)$ -module in the full-circle lax equalizer $\mathrm{CycSp}_{\mathbb{T}}^{(p)}$ can be dualizable only if (3.3) is an equivalence; together with dualizability of the underlying circle-equivariant A -module, this condition is also sufficient [KMGS21, Example 4.11].

The target of (3.3) vanishes. Indeed, by Lemma 3.2 and the tensor-ideal assertion in [NS18, Lemma I.3.8(ii)],

$$X^{tC_p} \simeq (A \otimes C)^{tC_p} \simeq 0. \quad (3.4)$$

On the other hand, extension of scalars gives

$$X \otimes_{A, \varphi_A} A^{tC_p} \simeq C \otimes A^{tC_p}. \quad (3.5)$$

After forgetting the residual circle action, the spectrum C is

$$C \simeq \Omega \Sigma_+^\infty S^1 \simeq \mathbb{S} \oplus \Sigma^{-1} \mathbb{S}.$$

A perfect field k of characteristic p is a perfectoid ring. Therefore [HN19, Addendum 1.3.3] identifies the p -complete cyclotomic Frobenius with an equivalence

$$\mathrm{THH}(k, \mathbb{Z}_p) \xrightarrow{\sim} \tau_{\geq 0}(\mathrm{THH}(k, \mathbb{Z}_p)^{tC_p}). \quad (3.6)$$

The spectrum $A = \mathrm{THH}(k)$ is a connective Hk -module and is therefore already p -complete. Thus (3.6) identifies A with the connective cover of A^{tC_p} . Since $\pi_0 A \cong k \neq 0$, the spectrum A^{tC_p} is nonzero. Hence (3.5) is nonzero, whereas (3.4) says that its target is zero. Therefore (3.3) is not an equivalence. The criterion of [KMGS21, Example 4.11] implies that $\rho_p Q$, and hence Q itself, is not dualizable. $\square$

Remark 3.5. *The failure in Proposition 3.4 is genuinely cyclotomic. The underlying A -module with circle action is dualizable: C has finite underlying spectrum and hence is dualizable in $\mathrm{Sp}^{B\mathbb{T}}$, and the free-module functor $A \otimes -$ is symmetric monoidal. Thus the only failed part of the Kondyrev–Mazel–Gee–Shah criterion is the Tate assembly equivalence.*

4. REALIZATION BY THE CYCLOTOMIC THH OF A MOTIVE

4.1. Cyclotomic THH on localizing motives. For a rigid small symmetric monoidal stable ∞ -category $\mathcal{E}$, Hoyo–Scherotzke–Sibilla construct a symmetric monoidal universal localizing invariant

$$U_{\mathcal{E}}^{\mathrm{loc}}: \mathrm{Cat}^{\mathrm{perf}}(\mathcal{E}) \longrightarrow \mathrm{Mot}_{\mathcal{E}}^{\mathrm{loc}}.$$

For every presentably symmetric monoidal stable ∞ -category $\mathcal{D}$, composition with $U_{\mathcal{E}}^{\mathrm{loc}}$ induces an equivalence between colimit-preserving symmetric monoidal functors $\mathrm{Mot}_{\mathcal{E}}^{\mathrm{loc}} \rightarrow \mathcal{D}$ and symmetric monoidal localizing invariants $\mathrm{Cat}^{\mathrm{perf}}(\mathcal{E}) \rightarrow \mathcal{D}$; see [HSS17, Theorems 5.17–5.18].

Comparison of the small and presentable base models. For an $\mathbb{E}_\infty$ -ring R , the presentable monoidal category Mod_R is rigid and compactly generated, with $(\mathrm{Mod}_R)^\omega = \mathrm{Perf}(R)$. Efimov’s comparison [Efi25, Subsections 1.5–1.6] consists of the equivalence

$$\mathrm{Ind}: \mathrm{Cat}^{\mathrm{perf}}(\mathrm{Perf}(R)) \xrightarrow{\sim} \mathrm{Cat}_{\mathrm{Mod}_R}^{\mathrm{cg}}$$

and the identification of the universal finitary localizing invariant on the right-hand side with the restriction of his invariant on dualizable Mod_R -modules. Consequently, the HSS category based on $\mathrm{Perf}(R)$ and Efimov’s category based on Mod_R are canonically equivalent as symmetric

monoidal presentable stable categories; we denote either model by $\mathrm{Mot}_R^{\mathrm{loc}}$. This identification is compatible with extension of scalars: for a map of $\mathbb{E}_\infty$ -rings $R \rightarrow S$, both constructions send a small R -linear category $\mathcal{C}$, after Ind-completion, to

$$\mathrm{Mod}_S \otimes_{\mathrm{Mod}_R} \mathrm{Ind}(\mathcal{C}).$$

We use this identification, and the same notation for the resulting base-change functors, throughout.

Topological Hochschild homology is a localizing invariant with values in integral cyclotomic spectra by the localization theorem of Blumberg–Mandell [BM12]; the statement in the Nikolaus–Scholze category is recorded in [McC24, Corollary 2.4.7]. It preserves filtered colimits and is symmetric monoidal as a functor

$$\mathrm{THH}: \mathrm{Cat}^{\mathrm{perf}} \longrightarrow \mathrm{CycSp}.$$

For an E_∞ -ring R , symmetric monoidality makes the THH of an R -linear category a module over $\mathrm{THH}(R)$ and gives a symmetric monoidal localizing invariant

$$\mathrm{THH}_R: \mathrm{Cat}^{\mathrm{perf}}(\mathrm{Perf}(R)) \longrightarrow \mathrm{Mod}_{\mathrm{THH}(R)}(\mathrm{CycSp}).$$

The module refinement is the formal consequence of this symmetric monoidality; its one-prime form is the module-valued cyclotomic THH used in [KMGS21, Example 4.11]. The universal property therefore gives colimit-preserving symmetric monoidal functors, denoted by the same symbols,

$$\begin{aligned} \mathrm{THH}: \mathrm{Mot}^{\mathrm{loc}} &\longrightarrow \mathrm{CycSp}, \\ \mathrm{THH}_R: \mathrm{Mot}_R^{\mathrm{loc}} &\longrightarrow \mathrm{Mod}_{\mathrm{THH}(R)}(\mathrm{CycSp}). \end{aligned} \tag{4.1}$$

The exact symmetric monoidal functor $\mathrm{Perf}(\mathbb{S}) \rightarrow \mathrm{Perf}(k)$ induces, by [HSS17, Theorem 5.18], a colimit-preserving symmetric monoidal base-change functor

$$b_k: \mathrm{Mot}^{\mathrm{loc}} \longrightarrow \mathrm{Mot}_k^{\mathrm{loc}}. \tag{4.2}$$

Lemma 4.1 (Base-change formula). *For every $N \in \mathrm{Mot}^{\mathrm{loc}}$, there is a natural equivalence of integral cyclotomic A -modules*

$$\mathrm{THH}_k(b_k N) \simeq A \otimes \mathrm{THH}(N).$$

Proof. Both sides define colimit-preserving symmetric monoidal functors

$$\mathrm{Mot}^{\mathrm{loc}} \longrightarrow \mathrm{Mod}_A(\mathrm{CycSp}).$$

By the universal property of $\mathrm{Mot}^{\mathrm{loc}}$, it is enough to compare them after composition with $U^{\mathrm{loc}}: \mathrm{Cat}^{\mathrm{perf}} \rightarrow \mathrm{Mot}^{\mathrm{loc}}$. For $\mathcal{C} \in \mathrm{Cat}^{\mathrm{perf}}$, symmetric monoidality of cyclotomic THH gives

$$\begin{aligned} \mathrm{THH}_k(b_k U^{\mathrm{loc}}(\mathcal{C})) &\simeq \mathrm{THH}(\mathrm{Perf}(k) \otimes \mathcal{C}) \\ &\simeq \mathrm{THH}(k) \otimes \mathrm{THH}(\mathcal{C}) \\ &= A \otimes \mathrm{THH}(U^{\mathrm{loc}}(\mathcal{C})). \end{aligned}$$

The equivalence is A -linear and cyclotomic. Universality now supplies the asserted natural equivalence on all motives. $\square$

4.2. Ramzi's universal THH-motive. Let

$$U: \text{CycSp} \longrightarrow \text{Sp}^{B\mathbb{T}}$$

denote the forgetful functor.

Lemma 4.2 (Unique cyclotomic structure on Tate-acyclic circle spectra). *Let $Z \in \text{Sp}^{B\mathbb{T}}$ satisfy $Z^{tC_\ell} \simeq 0$ for every prime ℓ . Then Z admits a unique cyclotomic structure up to a contractible space of choices, namely the one with all Frobenius maps zero. Moreover, if Z and Z' both have this property, every $\mathbb{T}$ -equivariant map $Z \rightarrow Z'$ promotes uniquely to a map of cyclotomic spectra.*

Proof. In the Nikolaus–Scholze definition, a cyclotomic structure on Z is precisely the collection of $\mathbb{T}$ -equivariant maps $\varphi_\ell: Z \rightarrow Z^{tC_\ell}$, one for each prime ℓ ; no further compatibility data are imposed [NS18, Definition II.1.1]. If every target is zero, each relevant mapping space is contractible. The compatibility square defining a map of cyclotomic spectra is then automatically commutative, again through a contractible space of choices. $\square$

We now invoke the calculation that supplies the motive to which the preceding lemma will be applied.

Proposition 4.3 (Universal THH-motive; Ramzi). *There exists an object $M_0 \in \text{Mot}^{\text{loc}}$ and a $\mathbb{T}$ -equivariant equivalence*

$$U \text{THH}(M_0) \simeq \text{coind}_1(\mathbb{S}) \simeq \Omega \Sigma_+^\infty \mathbb{T}. \quad (4.3)$$

With its intrinsic cyclotomic structure, $\text{THH}(M_0)$ is equivalent in CycSp to the object C of (3.1).

Proof. Efimov's self-duality, in the form recalled in [Ram26, Theorem 1.1], is the equivalence

$$\Psi: \text{Mot}^{\text{loc}} \xrightarrow{\sim} \text{Fun}^L(\text{Mot}^{\text{loc}}, \text{Sp}), \quad N \longmapsto K(N \otimes -).$$

Let M_0 be the object corresponding under Ψ to the localizing invariant THH . The composite of cyclotomic THH with the forgetful functor U is the usual circle-equivariant THH localizing invariant, by its construction on Cat^{perf} . Ramzi's calculation of the universal THH -motive, based on the S^1 -equivariant Day-convolution equivalence of [Ram25, Corollary 2.2] and the evaluation equivalence of [Ram25, Corollary 1.14], gives

$$U \text{THH}(M_0) \simeq \Omega \Sigma_+^\infty \mathbb{T}$$

as a $\mathbb{T}$ -spectrum; see [Ram26, Remark 2.4]. Equivariant Atiyah duality identifies the right-hand side with $\text{coind}_1(\mathbb{S})$, which proves (4.3).

By Lemma 3.2, the right-hand side of (4.3) is C_ℓ -Tate-acyclic for every prime ℓ . Therefore the underlying circle spectrum of $\text{THH}(M_0)$ is also Tate-acyclic. By Lemma 4.2, its intrinsic cyclotomic structure is uniquely the zero-Frobenius structure, and the equivalence (4.3) promotes uniquely to an equivalence in CycSp . By Lemma 3.1, the cyclotomic spectrum with this underlying circle spectrum and zero Frobenius maps is exactly C . $\square$

Remark 4.4. *Only the single intrinsic $\mathbb{T}$ -action in the cited calculation is used here. The possible $\mathbb{T} \times \mathbb{T}$ refinement mentioned in [Ram25, Remark 2.3] is not needed for the cyclotomic lifting argument.*

Define the promised k -linear motive by

$$M_k = b_k(M_0) \in \text{Mot}_k^{\text{loc}}. \quad (4.4)$$

By the preceding comparison, this is also the object denoted M_{Hk} when $\text{Mot}_k^{\text{loc}}$ and its base-change functors are written in Efimov's presentable-base formalism.

Corollary 4.5 (Cyclotomic realization). *There is an equivalence in $\mathrm{Mod}_A(\mathrm{CycSp})$*

$$\mathrm{THH}_k(M_k) \simeq Q.$$

Proof. By Lemma 4.1 and Proposition 4.3 and the definition of Q ,

$$\mathrm{THH}_k(M_k) = \mathrm{THH}_k(b_k M_0) \simeq A \otimes \mathrm{THH}(M_0) \simeq A \otimes C = Q.$$

□

5. PERFECT FIELDS OF POSITIVE CHARACTERISTIC

We can now combine the preceding constructions with Efimov's rigidity theorem.

Theorem 5.1 (Efimov). *The symmetric monoidal presentable stable ∞ -category $\mathrm{Mot}_k^{\mathrm{loc}}$ is rigid. In particular,*

$$(\mathrm{Mot}_k^{\mathrm{loc}})^\omega = (\mathrm{Mot}_k^{\mathrm{loc}})^{\mathrm{dual}}.$$

Proof. Apply [Efi25, Theorem 0.3] to the rigid presentable base Mod_k . In the ring-spectrum notation used here, the conclusion, including the displayed equality, is stated explicitly in [Ram26, Theorem 1.1]. □

Proof of Theorem 1.2. Apply Proposition 2.3 with

$$\begin{aligned} \mathcal{C} &= \mathrm{Mot}_k^{\mathrm{loc}}, \\ \mathcal{D} &= \mathrm{Mod}_A(\mathrm{CycSp}), \\ F &= \mathrm{THH}_k, \\ M &= M_k. \end{aligned}$$

The functor THH_k is colimit-preserving and symmetric monoidal by (4.1). The equality $\mathcal{C}^\omega = \mathcal{C}^{\mathrm{dual}}$ is Theorem 5.1. Finally, Corollary 4.5, Lemma 3.3, and Proposition 3.4 show that

$$F(M_k) \simeq Q$$

is compact but not dualizable. The abstract obstruction therefore gives

$$M_k \notin \mathrm{Loc}_{\mathrm{Mot}_k^{\mathrm{loc}}}((\mathrm{Mot}_k^{\mathrm{loc}})^\omega).$$

In particular, the compact objects do not generate $\mathrm{Mot}_k^{\mathrm{loc}}$, so $\mathrm{Mot}_k^{\mathrm{loc}}$ is not compactly generated. □

6. GENERAL CONNECTIVE $\mathbb{E}_2$ -RINGS

We now pass from the perfect-field theorem to an arbitrary nonzero connective $\mathbb{E}_2$ -ring. It is important here to descend a *specified* object. Noncompact generation by itself need not be reflected by extension of scalars. By contrast, the universal motive M_0 of Proposition 4.3 gives a compatible family of witnesses, and Ramzi's base-change formalism carries the compact envelope over the source into the compact envelope over the target.

6.1. Base-change and the universal witness.

Lemma 6.1 (Base-change and compact envelopes). *Let $f: R \rightarrow S$ be a map of $\mathbb{E}_2$ -ring spectra. The induced base-change functor*

$$f_*: \mathrm{Mot}_R^{\mathrm{loc}} \longrightarrow \mathrm{Mot}_S^{\mathrm{loc}}$$

sends compact objects to compact objects and satisfies

$$f_*\left(\mathrm{Loc}_{\mathrm{Mot}_R^{\mathrm{loc}}}((\mathrm{Mot}_R^{\mathrm{loc}})^\omega)\right) \subseteq \mathrm{Loc}_{\mathrm{Mot}_S^{\mathrm{loc}}}((\mathrm{Mot}_S^{\mathrm{loc}})^\omega).$$

Proof. The map f induces a continuous monoidal functor $\text{Mod}_R \rightarrow \text{Mod}_S$ between rigid monoidal bases. By [Efi25, Proposition 5.1], the resulting functor f_* on localizing motives is strongly continuous, or equivalently an internal left adjoint. This is also the base-change property used in [Ram26, Proof of the main theorem]. Thus its right adjoint f^* preserves colimits. If $P \in (\text{Mot}_R^{\text{loc}})^\omega$ and (N_i) is a filtered diagram in $\text{Mot}_S^{\text{loc}}$, then

$$\begin{aligned} \text{Map}_{\text{Mot}_S^{\text{loc}}} \left(f_* P, \text{colim}_i N_i \right) &\simeq \text{Map}_{\text{Mot}_R^{\text{loc}}} \left(P, f^* \text{colim}_i N_i \right) \\ &\simeq \text{Map}_{\text{Mot}_R^{\text{loc}}} \left(P, \text{colim}_i f^* N_i \right) \\ &\simeq \text{colim}_i \text{Map}_{\text{Mot}_S^{\text{loc}}} (f_* P, N_i). \end{aligned}$$

Hence $f_* P$ is compact. Put

$$\mathcal{L}_S = \text{Loc}_{\text{Mot}_S^{\text{loc}}}((\text{Mot}_S^{\text{loc}})^\omega).$$

Because f_* is exact and preserves colimits, $f_*^{-1}(\mathcal{L}_S)$ is a localizing subcategory of $\text{Mot}_R^{\text{loc}}$. The compactness assertion just proved gives $(\text{Mot}_R^{\text{loc}})^\omega \subseteq f_*^{-1}(\mathcal{L}_S)$. Therefore

$$\text{Loc}_{\text{Mot}_R^{\text{loc}}}((\text{Mot}_R^{\text{loc}})^\omega) \subseteq f_*^{-1}(\mathcal{L}_S),$$

which is equivalent to the asserted inclusion. $\square$

Lemma 6.2 (Compatible universal witnesses). *For every $\mathbb{E}_2$ -ring R , let*

$$b_R: \text{Mot}^{\text{loc}} \longrightarrow \text{Mot}_R^{\text{loc}}$$

denote extension of scalars and put $M_R = b_R(M_0)$. For every map $f: R \rightarrow S$ of $\mathbb{E}_2$ -rings, there is a natural equivalence

$$f_* M_R \simeq M_S.$$

Proof. Extension of scalars is functorial in the base ring. More precisely, the composite

$$\text{Mot}^{\text{loc}} \xrightarrow{b_R} \text{Mot}_R^{\text{loc}} \xrightarrow{f_*} \text{Mot}_S^{\text{loc}}$$

is naturally equivalent to extension of scalars along the composite $\mathbb{S} \rightarrow R \rightarrow S$, namely b_S . Evaluation at M_0 gives the claim. $\square$

When $R = Hk$ for a field k , the comparison in Section 4 identifies this object with the motive M_k of (4.4).

Proposition 6.3 (Descent of the distinguished obstruction). *Let $f: R \rightarrow S$ be a map of $\mathbb{E}_2$ -rings. If*

$$M_S \notin \text{Loc}_{\text{Mot}_S^{\text{loc}}}((\text{Mot}_S^{\text{loc}})^\omega),$$

then

$$M_R \notin \text{Loc}_{\text{Mot}_R^{\text{loc}}}((\text{Mot}_R^{\text{loc}})^\omega).$$

Proof. If M_R belonged to the compact envelope over R , then Lemma 6.1 would place $f_* M_R$ in the compact envelope over S . By Lemma 6.2, this object is M_S , contradicting the hypothesis. $\square$

6.2. The residue-field reduction.

Lemma 6.4 (A perfect residue field in the rationally zero case). *Let R be a nonzero connective $\mathbb{E}_2$ -ring such that $R_{\mathbb{Q}} = R \otimes H\mathbb{Q}$ vanishes. Then there exist a prime p , a perfect field k of characteristic p , and a map of $\mathbb{E}_2$ -rings*

$$R \longrightarrow Hk.$$

Proof. Set $A = \pi_0 R$. Since R is an $\mathbb{E}_2$ -ring, A is a commutative ring. It is nonzero: if $1 = 0$ in A , then the unit acts as both zero and the identity on every $\pi_i R$, so all homotopy groups of R vanish. Choose a maximal ideal $\mathfrak{m} \subset A$ and write $\kappa = A/\mathfrak{m}$.

The residue field κ has positive characteristic. Indeed, if $\text{char}(\kappa) = 0$, then every nonzero integer maps to a unit in κ . The quotient map $A \rightarrow \kappa$ would therefore extend uniquely to a unital map

$$A \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \kappa.$$

In particular, $A \otimes \mathbb{Q}$ would be nonzero. On the other hand, rational localization is flat, and connectivity gives

$$\pi_0(R_{\mathbb{Q}}) \cong \pi_0 R \otimes_{\mathbb{Z}} \mathbb{Q} = A \otimes_{\mathbb{Z}} \mathbb{Q} = 0,$$

a contradiction.

Let k be a perfect closure of κ . The canonical zeroth Postnikov truncation in $\text{Alg}_{\mathbb{E}_2}(\text{Sp}^{\text{cn}})$ is a map of $\mathbb{E}_2$ -rings $R \rightarrow HA$. Composing it with the $\mathbb{E}_{\infty}$ -maps induced by $A \rightarrow \kappa \rightarrow k$ gives

$$R \longrightarrow HA \longrightarrow H\kappa \longrightarrow Hk,$$

as required. $\square$

Corollary 6.5 (The rationally zero branch). *Let R be a nonzero connective $\mathbb{E}_2$ -ring with $R_{\mathbb{Q}} = 0$. Then the distinguished motive $M_R = b_R(M_0)$ satisfies*

$$M_R \notin \text{Loc}_{\text{Mot}_R^{\text{loc}}}((\text{Mot}_R^{\text{loc}})^{\omega}).$$

Proof. Choose a perfect field k of positive characteristic and a map $f: R \rightarrow Hk$ as in Lemma 6.4. By Theorem 1.2, M_k lies outside the compact envelope of $\text{Mot}_k^{\text{loc}}$. The claim now follows from Proposition 6.3. $\square$

Proof of Theorem 1.1. Let R be a nonzero connective $\mathbb{E}_2$ -ring. If $R_{\mathbb{Q}} = R \otimes H\mathbb{Q}$ is nonzero, the conclusion is the main theorem of [Ram26]. If $R_{\mathbb{Q}} = 0$, the conclusion is Corollary 6.5. These two cases are exhaustive. $\square$

Remark 6.6 (The witness and the role of connectivity). *In the rationally zero branch, the witness M_R is not asserted to be compact. Compactness is used only for its perfect-field image under relative cyclotomic THH, while ρ_p is used only to detect failure of dualizability. Connectivity enters the general theorem through the canonical map $R \rightarrow H\pi_0 R$; no assertion is made here for nonconnective $\mathbb{E}_2$ -rings.*

Remark 6.7 (Prime-supported examples). *Perfectness is required only after passing to the detecting residue field. Thus Corollary 6.5 applies, for example, to every field of positive characteristic, whether or not it is perfect, and to every nonzero connective $\mathbb{E}_2$ -algebra over $H\mathbb{F}_p$. In all these cases it identifies the functorial object $M_R = b_R(M_0)$, rather than merely an unspecified motive, outside the compact envelope.*

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